Case Study: Member Attendance & Fines Analytics
Author: Okikioluwa DASHO
Organization: Just us Students Organization
Dataset Scope: 73 Members | 11 Meeting Days

1. Executive Summary & The Problem

In student organizations and academic associations, tracking meeting attendance, lateness, and associated fines is frequently handled through manual paper sheets or messy, unstandardized spreadsheets. For this organization, this manual approach resulted in opacity, uncollected disciplinary fines, and difficulty in identifying engagement trends across members.

This case study outlines my first end-to-end data analyst project, where I transformed raw attendance logs into a structured, relational dataset, built pivot-based analytical models, and uncovered actionable behavioral and financial insights.

2. Approach & Methodology

To turn unstructured records into an analytical engine, I executed a structured data pipeline:

- Data Cleaning & Standardization: Standardized raw attendance inputs using uniform status codes: P (Present), L (Late), E (Excused absence), and U (Unexcused absence).
- Power Query Transformation (Long Format): Transformed wide-format attendance tables into a clean, normalized long format (one row per member per meeting date) using Power Query (Unpivot Columns). This relational structure served as the foundation for all pivot tables and charts.
- Pivot-Based Multidimensional Analysis: Developed dedicated analytical modules for Attendance, Fines, and Demographics to compute collection rates, fine distributions, and category breakdowns.
- Statistical Modeling & Outlier Detection: Applied statistical functions (AVERAGE, MEDIAN, MODE, STDEV.S) to evaluate distribution shapes and implemented an objective outlier detection rule.
- Chi-Square Contingency Analysis: Performed a Chi-Square test of independence to test whether unexcused absences are statistically associated with members gender.

3. Key Findings

Finding 1: Attendance Distribution & Outlier Identification

- Distribution Shape: The attendance data exhibited a left-skewed distribution where Mean (73%) < Median (80%) < Mode (91%)}, indicating that a typical member attended over 73% of scheduled meetings.
- Temporal Dips & Peaks: Meeting-level analysis revealed significant variance in engagement. For instance, June 28th experienced a low attendance rate of 64%, driven by unusual absences, whereas June 21st and May 10th sessions peaked at 83%.
- Outlier Tracking: Members whose attendance rates fell significantly below the norm were successfully flagged for targeted administrative follow-up.

Finding 2: Financial Collections & Top Owing Members

- Revenue vs. Outstanding Dues: Financial aggregation revealed a major collection backlog. While 16 members had clean records (not owing fines) and 23 members settled their dues, 34 members had outstanding balances.

- Monetary Breakdown:
  - Total collected lateness fines: ₦3,900 vs. Unpaid lateness: ₦15,600.
  - Total collected unexcused absence fines: ₦23,000 vs. Unpaid unexcused fines: ₦61,500.

- Top Debtors: The analysis isolated specific repeat offenders. For lateness fines with no excuses, top members included David (₦1,500) ,Funmi (₦1,200) and Phebe(₦900). For unexcused absences, top owing members were led by Raham (₦4,000), Daniel (₦3,000) and Tobi(₦2,500).

Finding 3: Gender Segmentation Insights & Chi-Square Analysis

- Cross-tabulating demographic segmentation against attendance and fines revealed clear behavioral contrasts:

1. Female members maintained a higher average attendance rate (77%) and lower unexcused absence rates (12%) compared to male members (70% attendance and 19% unexcused absence).
2. Chi-Square Test Results: A contingency analysis yielded a p-value of 0.0419, which is below the standard 0.05 significance threshold. This statistically confirms that unexcused absences are dependent on gender, providing leadership with proof for targeted engagement strategies.

4. What I Would Improve With More Time

- Automated Reminders & Notifications: Integrate the dataset with an email or messaging script to automatically dispatch fine payment reminders to members exceeding specific debt thresholds.

- Interactive Dashboarding: Export the cleaned relational model into Power BI or Looker Studio to build an executive, filterable dashboard with dynamic date slicers.

- Longitudinal Tracking: Expand the model across multiple academic quarters (Quarter-over-Quarter analysis) to measure whether administrative interventions successfully reduce chronic lateness and unexcused absences over time.

Employer Takeaway:
This project proves my ability to take messy, real-world operational logs, engineer clean relational tables, apply statistical tests, and translate numbers into strategic business and organizational recommendations.